

Physics 514, Problem Set 6: Stress Tensors & Symmetries

Due Tuesday, March 8 at Noon

Please place your solutions in the box outside my office before the due date. You are encouraged to discuss these problems with your colleagues, but you must write up your own solutions; the solutions you hand in should reflect your own work and understanding.

Reading: Chapters 1, 2, 3, & 4 of Carroll.

1. Consider electromagnetism in flat Minkowski space. The electromagnetic potential A_μ is the one-form with components $(-V, \mathbf{A})$ in Cartesian coordinates, where V is the electric potential and $\mathbf{A}$ is the vector potential. The electromagnetic fieldstrength $F_{\mu\nu}$ is the two form $F = dA$.

- (a) Show that in Minkowski space, Maxwell's equations can be written as

$$dF = 0, \quad \partial_\mu F^{\mu\nu} = j^\nu$$

where j^μ is the 4-vector with components $(\rho, \mathbf{j})$, where ρ and $\mathbf{j}$ are charge and current densities, respectively. Show that the second of these equations implies that the current 4-vector j^μ obeys

$$\partial_\mu j^\mu = 0.$$

A vector obeying this equation is known as a *conserved current*. Check that this equation is equivalent to the continuity equation for charge density.

- (b) Show that these equations of motion can be derived from the action

$$\int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu j^\mu \right)$$

- (c) Using the principle of minimal coupling, write down the action and equations of motion of electromagnetism in curved space. The current 4-vector now obeys

$$\nabla_\mu j^\mu = 0$$

This is the formula for a conserved current – i.e. the continuity equation – in curved space-time.

- (d) Consider the case where $j^\mu = 0$. Compute the stress tensor of electromagnetism by varying the action you derived in the previous part with respect to the metric. Check that the equations of motion imply that this stress tensor is conserved, $\nabla^\mu T_{\mu\nu} = 0$.

2. Given a vector v^μ and a tensor $T_{\mu\nu}$ we define the Lie derivative of T with respect to v to be

$$\mathcal{L}_v T_{\mu\nu} = v^\alpha \partial_\alpha T_{\mu\nu} + \partial_\mu v^\alpha T_{\alpha\nu} + \partial_\nu v^\alpha T_{\mu\alpha}$$

- (a) Check that this is a tensor of rank $(0, 2)$.
- (b) Show that in the above formula for the Lie derivative we can replace the partial derivative by the covariant derivative and the result would be unchanged.
3. Assume that we have a metric $g_{\mu\nu}$ with a Killing vector k^μ . As explained in class, this means that the vector k^μ obeys $\mathcal{L}_k g_{\mu\nu} = \nabla_{(\mu} k_{\nu)} = 0$. This vector k^μ generates a continuous symmetry of the geometry.

- (a) Assume that $T_{\mu\nu}$ is the stress tensor of some matter system. Define $j_\mu = T_{\mu\nu} k^\nu$. Show that $\nabla_\mu j^\mu = 0$. That means that the components of j^μ obey a continuity equation. Thus when the geometry has a timelike (or spacelike) Killing symmetry we can define local energy (or momentum) density from a stress tensor.
- (b) Imagine that we have a particle moving along a geodesic $x^\mu(\lambda)$. Take λ to be an affine parameter. Let $v^\mu = dx^\mu/d\lambda$ be the tangent vector to the geodesic. Define

$$I(\lambda) = k_\mu v^\mu$$

This is a function of points on the worldline. Show that I is a constant along the geodesic. The quantity I is the Noether charge associated with symmetry generated by k^μ .

- (c) Consider a particle moving along a timelike geodesic in Minkowski space. Show that when $k = \partial_t$ the constant I defined above is proportional to the energy of the particle in special relativity. Show that when $k = \partial_i$ the constant I is proportional to the momentum of the particle in the x^i direction.
4. In this problem we again take k^μ to be a Killing vector for a geometry with metric $g_{\mu\nu}$.
- (a) Show that $\nabla_\mu \nabla_\nu k^\rho = R^\rho{}_{\nu\mu\sigma} k^\sigma$.
- (b) Show that $k^\mu \nabla_\mu R = 0$. (*Hint: choose a coordinate y such that $k^\mu \nabla_\mu = \frac{\partial}{\partial y}$ and the metric is independent of y . Argue that R must be independent of y).*
- (c) Show that the commutator of two Killing vectors is a Killing vector.
- (d) Assume that the metric $g_{\mu\nu}$ solves the vacuum Einstein equations and possesses a Killing vector k^μ . Show that $A_\mu = k_\mu$ is a solution of Maxwell's equation in this curved spacetime.

Problem 1

We define:

$$A_\mu = (-V, \vec{A}) \quad (1)$$

and:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (2)$$

(ignoring prefactors)

Let's start with $dF = 0$. In index notation,

$$\partial_\alpha F_{\mu\nu} + \partial_\mu F_{\nu\alpha} + \partial_\nu F_{\alpha\mu} = 0$$

where I have used the antisymmetry of F .

Letting $\mu=0$, $\nu=i$, $\alpha=j$ & swapping indices,

$$\partial_j F_{0i} + \partial_0 F_{ij} - \partial_i F_{0j} = 0$$

$$\Rightarrow \partial_i E_j - \partial_j E_i = -\partial_t \tilde{\epsilon}_{ijk} B^k$$

assume $i \neq j$.
If it is, then
it is trivial.

This is precisely the statement:

$$\nabla \times \vec{E} = -\partial_t \vec{B}$$

Note that we have used the fact that (2) is identical to the matrix:

$$F_{\mu\nu} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix}$$

$$\begin{aligned} E_i &= F_{i0} \\ F_{ij} &= \tilde{\epsilon}_{ijk} B^k \end{aligned}$$

It's easy to check $\mu = \nu = \alpha = 0$ just vanishes, as does $\mu = \alpha = 0, \nu = i$.

Next let α, μ, ν be i, j, k , where all three are different. Then:

$$\partial_i \tilde{\epsilon}_{jka} B^a + \partial_j \tilde{\epsilon}_{kia} B^a + \partial_k \tilde{\epsilon}_{ija} B^a = 0. \quad (3)$$

$$\Rightarrow \partial_i B_i + \partial_j B_j + \partial_k B_k = 0$$

which is precisely $\nabla \cdot \vec{B} = 0$.

It's easy to see that (3) would be trivial if the indices weren't all different.

The cases $\alpha = i, \mu = \nu = 0$, or with $\nu = i$ simply vanish.

Every other variation of (2) just restates the cases we have covered. So we have shown that:

$$dF = 0 \iff \begin{aligned} \nabla \times \vec{E} &= -\partial_t \vec{B} \\ \nabla \cdot \vec{B} &= 0. \end{aligned}$$

(question is wrong...)

Next consider $\partial_\mu F^{\nu\mu} = j^\nu$ (4)

When $v=0$ this states:

$$\partial_1 E_1 + \partial_2 E_2 + \partial_3 E_3 = \rho$$

which is the statement $\nabla \cdot \vec{E} = \rho/\epsilon_0$ in natural units. Note that we have used that:

$$F^{i0} = -F_{i0}$$

since $F^{\mu\nu} = \eta^{\mu\alpha} \eta^{\nu\beta} F_{\alpha\beta}$ & so similarly,

$$F^{\dot{i}\dot{j}} = F_{\dot{i}\dot{j}}$$

So $v=1$ gives:

$$-\partial_t E_1 + \partial_2 B_3 - \partial_3 B_2 = j^1$$

$$\Rightarrow (\nabla \times \vec{B})_1 = j^1 + \partial_t E_1$$

Including the other indices, this becomes the statement:

$$\nabla \times \vec{B} = \vec{j} + \partial_t \vec{E}$$

which is the final Maxwell eqn. in natural units. So we have shown:

$$\partial_\mu F^{\nu\mu} = j^\nu \iff \begin{aligned} \nabla \cdot \vec{E} &= \rho/\epsilon_0 \\ \nabla \times \vec{B} &= \mu_0 \vec{j} + \frac{1}{c^2} \partial_t \vec{E} \end{aligned}$$

From (4):

$$\partial_\nu j^\nu = \partial_\nu \partial_\mu F^{\nu\mu}$$

But ∂_μ 's commute & F is antisymmetric, so:

$$\partial_\nu \partial_\mu F^{\nu\mu} = -\partial_\mu \partial_\nu F^{\mu\nu}$$

$$\Rightarrow \partial_\nu \partial_\mu F^{\nu\mu} = -\partial_\nu \partial_\mu F^{\nu\mu}$$

$$\Rightarrow \partial_\nu \partial_\mu F^{\nu\mu} = 0$$

$$\Rightarrow \partial_\nu j^\nu = 0.$$

Expanding it gives:

$$\partial_t \rho + \nabla \cdot \vec{j} = 0$$

which is the continuity equation.

b. Consider the action:

$$S = \int d^4x \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu j^\mu \right)$$

As usual we get the EOMs by:

$$\frac{\partial \mathcal{L}}{\partial A_\nu} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} \right) = 0. \quad (5)$$

Evidently:

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = j^\nu \quad (6)$$

On the other hand, using (2),

$$\frac{\partial \mathcal{L}}{\partial (\partial_\mu A_\nu)} = -\frac{1}{4} \left\{ F^{\alpha\beta} \frac{\partial}{\partial (\partial_\mu A_\nu)} (F_{\alpha\beta}) + F_{\alpha\beta} \frac{\partial}{\partial (\partial_\mu A_\nu)} (F^{\alpha\beta}) \right\}$$

$$= -\frac{1}{4} \left\{ F^{\alpha\beta} (\delta_\alpha^\mu \delta_\beta^\nu - \delta_\beta^\mu \delta_\alpha^\nu) + F_{\alpha\beta} \eta^{\alpha\sigma} \eta^{\beta\rho} (\delta_\sigma^\mu \delta_\rho^\nu - \delta_\rho^\mu \delta_\sigma^\nu) \right\}$$

$$= F^{\nu\mu}, \quad \text{using antisymmetry of } F.$$

With (5) & (6) we get:

$$\partial_\mu F^{\nu\mu} = j^\nu$$

Since the statement $dF = 0$ is the same as $d^2A = 0$, & since this is true for any A , there is nothing to "derive" here.

So indeed, this action recovers Maxwell's eqns.

c. The action is now:

$$S = \int d^4x \sqrt{-g} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_\mu j^\mu \right) \quad (7)$$

where:

$$F_{\mu\nu} = \nabla_\mu A_\nu - \nabla_\nu A_\mu \quad (8)$$

The EOMs become:

$$dF = 0 \quad (9)$$

$$\nabla_\mu F^{\nu\mu} = 0 \quad (10)$$

d. We let $g^{\mu\nu} \rightarrow g^{\mu\nu} + \delta g^{\mu\nu}$ to get:

$$\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}$$

$$\begin{aligned}\delta(F_{\mu\nu}F^{\mu\nu}) &= (g^{\alpha\mu}\delta g^{\beta\nu} + g^{\beta\nu}\delta g^{\alpha\mu})F_{\mu\nu}F_{\alpha\beta} \\ &= 2g^{\alpha\beta}\delta g^{\mu\nu}F_{\beta\nu}F_{\alpha\mu}\end{aligned}$$

$$\delta(A_{\mu}j^{\mu}) = A_{\mu}j_{\nu}\delta g^{\mu\nu}$$

So that:

$$\begin{aligned}\delta S &= \int d^4x \left(-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta} + A_{\alpha}j^{\alpha} \right) \left(-\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu} \right) \\ &\quad + \int d^4x \sqrt{-g} \left(-\frac{1}{2}g^{\alpha\beta}F_{\beta\nu}F_{\alpha\mu} + A_{\mu}j_{\nu} \right) \delta g^{\mu\nu}\end{aligned}$$

So :

$$\begin{aligned}T_{\mu\nu} &= -\frac{2}{\sqrt{-g}}\frac{\delta S}{\delta g^{\mu\nu}} \\ &= g_{\mu\nu}\left(-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta} + A_{\alpha}j^{\alpha}\right) + g^{\alpha\beta}F_{\beta\nu}F_{\alpha\mu} - 2A_{\mu}j_{\nu}\end{aligned}$$

Setting $j^{\mu} = 0$,

$$T_{\mu\nu} = -\frac{1}{4}g_{\mu\nu}F_{\alpha\beta}F^{\alpha\beta} + g^{\alpha\beta}F_{\beta\nu}F_{\alpha\mu}$$

(from (10), it is clear that $\nabla^{\mu}T_{\mu\nu} = 0$)

Problem 2

a. We aim to show:

$$\mathcal{L}_v T_{\mu'v'} = \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \mathcal{L}_v T_{\mu\nu}$$

Well,

$$\mathcal{L}_v T_{\mu'v'} = v^{\alpha'} \partial_{\alpha'} T_{\mu'v'} + T_{\alpha'v'} \partial_{\mu'} v^{\alpha'} + T_{\mu'\alpha'} \partial_{v'} v^{\alpha'}$$

$$= \frac{\partial x^{\alpha'}}{\partial x^\beta} v^\beta \frac{\partial x^\alpha}{\partial x^{\alpha'}} \partial_\alpha \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} T_{\mu\nu} \right)$$

$$+ \frac{\partial x^\beta}{\partial x^{\alpha'}} \frac{\partial x^\nu}{\partial x^{\nu'}} T_{\beta\nu} \frac{\partial x^\mu}{\partial x^{\mu'}} \partial_\mu \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha} v^\alpha \right)$$

$$+ \frac{\partial x^\beta}{\partial x^{\alpha'}} \frac{\partial x^\mu}{\partial x^{\mu'}} T_{\mu\beta} \frac{\partial x^\nu}{\partial x^{\nu'}} \partial_{v'} \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha} v^\alpha \right)$$

$$= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \mathcal{L}_v T_{\mu\nu} + T_{\mu\nu} v^{\alpha'} \partial_{\alpha'} \left(\frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \right)$$

$$+ \frac{\partial x^\beta}{\partial x^{\alpha'}} \frac{\partial x^\nu}{\partial x^{\nu'}} T_{\beta\nu} v^{\alpha'} \partial_{\mu'} \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha} \right)$$

$$+ \frac{\partial x^\beta}{\partial x^{\alpha'}} \frac{\partial x^\mu}{\partial x^{\mu'}} T_{\beta\mu} v^{\alpha'} \partial_{v'} \left(\frac{\partial x^{\alpha'}}{\partial x^\alpha} \right)$$

$$= \frac{\partial x^\mu}{\partial x^{\mu'}} \frac{\partial x^\nu}{\partial x^{\nu'}} \mathcal{L}_v T_{\mu\nu} .$$

In the penultimate step I have just used the product rule to expand & isolate terms, then simplified using the chain rule. In the final step, the extra terms yield zero since:

$$\begin{aligned}\partial_{\mu'} \left(\frac{\partial x^{\alpha'}}{\partial x^{\alpha}} \right) &= \partial_{\alpha} \left(\frac{\partial x^{\alpha'}}{\partial x^{\mu'}} \right) \\ &= \partial_{\alpha} (\delta_{\mu'}^{\alpha'}) \\ &= 0.\end{aligned}$$

b. Let:

$$\mathcal{L}'_v T_{\mu\nu} = v^{\alpha} \nabla_{\alpha} T_{\mu\nu} + T_{\alpha\nu} \nabla_{\mu} v^{\alpha} + T_{\mu\alpha} \nabla_{\nu} v^{\alpha}$$

Then:

$$\begin{aligned}\mathcal{L}'_v T_{\mu\nu} &= \mathcal{L}_v T_{\mu\nu} - \underbrace{v^{\alpha} \Gamma_{\alpha\mu}^{\lambda} T_{\lambda\nu}} \\ &\quad - \underbrace{v^{\alpha} \Gamma_{\alpha\nu}^{\lambda} T_{\mu\lambda}} + \underbrace{T_{\alpha\nu} \Gamma_{\mu\lambda}^{\alpha} v^{\lambda}} \\ &\quad + \underbrace{T_{\mu\alpha} \Gamma_{\nu\lambda}^{\alpha} v^{\lambda}}\end{aligned}$$

$$= \mathcal{L}_v T_{\mu\nu}$$

(after relabeling dummy indices & using the symmetry of Γ)

Problem 3

a. Let: $j_\mu = T_{\mu\nu} k^\nu$

Then,

$$\nabla_\mu j^\mu = k_\nu \nabla_\mu T^{\mu\nu} + T^{\mu\nu} \nabla_\mu k_\nu$$

Now, T is a stress tensor so the first term vanishes. The second term can also be written as:

$$\begin{aligned} T^{\mu\nu} \nabla_\mu k_\nu &= T^{\nu\mu} \nabla_\mu k_\nu \\ &= T^{\mu\nu} \nabla_\nu k_\mu \\ &= \frac{1}{2} T^{\mu\nu} (\nabla_\mu k_\nu + \nabla_\nu k_\mu) \\ &= 0 \end{aligned}$$

where I used the symmetry of T , then relabelled dummy indices. The last line uses the definition of Killing vectors. So:

$$\nabla_\mu j^\mu = 0.$$

b. We want to show $I'(\lambda) = 0$. Well,

$$\begin{aligned}
\frac{d}{d\lambda} I &= v^\mu \partial_\mu I \\
&= v^\mu \nabla_\mu I \\
&= v^\mu v^\nu \nabla_\mu k_\nu + k_\nu v^\mu \nabla_\mu v^\nu \\
&= 0.
\end{aligned}$$

The first line is just the chain rule, the second follows from the definition of ∇ . In the final line, the first term is zero using the same argument as earlier, whereas the second term is just the geodesic equation, which evaluates to zero for affine parameters.

c. In SR, we define:

$$p^\mu = m v^\mu \quad \text{where the particle has 4-velocity } v^\mu.$$

with p^0 (p^i) representing energy (momentum).

Evidently, $k_\mu v^\mu$, if k is along one of the axes, picks out that component of v^μ .

So for example, if $k = \partial_t$, or equivalently $k^\mu = (1, 0, 0, 0)$, then $I = p^0/m$, which is proportional to the energy.

(same thing for momentum).

Problem 4

a. By definition:

$$\begin{aligned}\nabla_\mu \nabla_\nu k^\rho &= \partial_\mu (\partial_\nu k^\rho + \Gamma_{\nu\lambda}^\rho k^\lambda) \\ &\quad + \Gamma_{\mu\sigma}^\rho (\partial_\nu k^\sigma + \Gamma_{\nu\lambda}^\sigma k^\lambda) \\ &\quad - \Gamma_{\mu\nu}^\sigma (\partial_\sigma k^\rho + \Gamma_{\sigma\lambda}^\rho k^\lambda)\end{aligned}$$

Pick coor'ds for which k is a basis vector so that $\partial k = 0$ & $k^\mu \partial_\mu g_{\nu\sigma} = 0$. Then:

$$\begin{aligned}\nabla_\mu \nabla_\nu k^\rho &= k^\lambda \partial_\mu \Gamma_{\nu\lambda}^\rho + \Gamma_{\mu\sigma}^\rho \Gamma_{\nu\lambda}^\sigma k^\lambda \\ &\quad - \Gamma_{\mu\nu}^\sigma \Gamma_{\sigma\lambda}^\rho k^\lambda\end{aligned}$$

$$= (\partial_\mu \Gamma_{\nu\lambda}^\rho + \Gamma_{\mu\sigma}^\rho \Gamma_{\nu\lambda}^\sigma - \Gamma_{\mu\nu}^\sigma \Gamma_{\sigma\lambda}^\rho) k^\lambda$$

$$\begin{aligned}= &(\partial_\mu \Gamma_{\nu\lambda}^\rho - \partial_\lambda \Gamma_{\mu\nu}^\rho + \Gamma_{\mu\sigma}^\rho \Gamma_{\nu\lambda}^\sigma \\ &\quad - \Gamma_{\mu\nu}^\sigma \Gamma_{\sigma\lambda}^\rho) k^\lambda\end{aligned}$$

where I've added $-k^\lambda \partial_\lambda \Gamma_{\mu\nu}^\rho$ which is zero anyway. But by definition, this is:

$$\nabla_\mu \nabla_\nu k^\rho = R^\rho{}_{\nu\mu\lambda} k^\lambda$$

To be clear, this used a particular coord. system, but the expression is tensorial so it holds true generally.

b. By definition of covariant derivatives, $\nabla_\mu R = \partial_\mu R$, & we can again choose coords where k is a basis vector.

Then, $k^\mu \partial_\mu g_{\nu\sigma} = 0$. Or, the metric is independent of the coordinate represented by k . But R is only a function of the metric & its derivatives. So naturally:

$$k^\mu \partial_\mu R = 0$$

$$\Rightarrow k^\mu \nabla_\mu R = 0.$$

c. Take two Killing vectors v & w . Define:

$$k_\mu = [v, w]_\mu = v^\lambda \nabla_\lambda w_\mu - w^\lambda \nabla_\lambda v_\mu$$

So that:

$$\begin{aligned} \nabla_\mu k_\nu &= (\nabla_\mu v^\lambda)(\nabla_\lambda w_\nu) + v^\lambda \nabla_\mu \nabla_\lambda w_\nu \\ &\quad - (\nabla_\mu w^\lambda)(\nabla_\lambda v_\nu) - w^\lambda \nabla_\mu \nabla_\lambda v_\nu \end{aligned}$$

Or, using (a): (ignoring prefactors)

$$\begin{aligned}
\nabla_{(\mu} k_{\nu)} &= (\nabla_{\mu} v^{\lambda})(\nabla_{\lambda} w_{\nu}) - (\nabla_{\mu} w^{\lambda})(\nabla_{\lambda} v_{\nu}) \\
&\quad + \underbrace{v^{\lambda} R_{\nu\lambda\mu\alpha} w^{\alpha}} - \underbrace{w^{\lambda} R_{\nu\lambda\mu\alpha} v^{\alpha}} \\
&\quad + (\nabla_{\nu} v^{\lambda})(\nabla_{\lambda} w_{\mu}) - (\nabla_{\nu} w^{\lambda})(\nabla_{\lambda} v_{\mu}) \\
&\quad + \underbrace{v^{\lambda} R_{\mu\lambda\nu\alpha} w^{\alpha}} - \underbrace{w^{\lambda} R_{\mu\lambda\nu\alpha} v^{\alpha}}
\end{aligned}$$

where the color coded terms cancel because the Riemann tensor is symmetric under exchange of the first & last two indices.

Finally, we can use $\nabla_{\mu} w_{\nu} = -\nabla_{\nu} w_{\mu}$ to rewrite the other terms:

$$\begin{aligned}
\nabla_{(\mu} k_{\nu)} &= (\nabla_{\mu} v^{\lambda})(\nabla_{\lambda} w_{\nu}) + (\nabla_{\mu} w_{\lambda})(\nabla_{\nu} v^{\lambda}) \\
&\quad - (\nabla_{\nu} v^{\lambda})(\nabla_{\mu} w_{\lambda}) - (\nabla_{\lambda} w_{\nu})(\nabla_{\mu} v^{\lambda}) \\
&= 0.
\end{aligned}$$

So, k is a Killing vector.

d. Let's start by noting that we can't just set $\partial_{\mu} k_{\nu}$ to zero. We can set $\partial_{\mu} k^{\nu}$ to

zero, but $\partial_\mu k_\nu = \partial_\mu (g_{\nu\lambda} k^\lambda)$ & it's not clear we can set both to zero.

Anyway, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. But we can freely replace ∂ with ∇ . So:

$$\begin{aligned} F^{\mu\nu} &= \nabla^\mu k^\nu - \nabla^\nu k^\mu \\ &= 2\nabla^\mu k^\nu \end{aligned}$$

$$\Rightarrow \nabla_\nu F^{\mu\nu} = 2\nabla_\nu \nabla^\mu k^\nu$$

Using (a) again:

$$\begin{aligned} \nabla_\nu F^{\mu\nu} &= 2g^{\mu\alpha} R^\nu_{\alpha\nu\sigma} k^\sigma \\ &= 2g^{\mu\alpha} R_{\alpha\sigma} k^\sigma \end{aligned}$$

But in vacuum the Ricci tensor vanishes, so:

$$\nabla_\nu F^{\mu\nu} = 0.$$

So Maxwell's eqn.s are satisfied, since $d\overline{F} = 0$ is automatic.